Synopsis

Disease Prediction using Machine Learning

Project Id: 1

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- **Problem Description:**

With the rapid growth of healthcare data, predicting disease risk at an early stage has become extremely important for improving patient outcomes and reducing medical costs. Many diseases can be prevented or controlled if detected early using clinical indicators such as age, blood pressure, cholesterol levels, blood sugar, and lifestyle habits like smoking.

This project focuses on developing an end-to-end machine learning system to predict whether a person is at **high risk or low risk of disease**. The system processes raw healthcare data, performs data cleaning and transformation, and applies feature selection techniques to improve model performance.

The dataset includes attributes such as:

- Age
- Gender
- Blood Pressure (Systolic & Diastolic)
- Smoking Habit
- Other clinical features

The target variable **Disease_Risk** is converted into binary form:

- Low / Medium → 0
- High → 1

This system helps doctors and healthcare professionals in:

- Early diagnosis
- Risk assessment
- Preventive healthcare planning

- **Architecture**

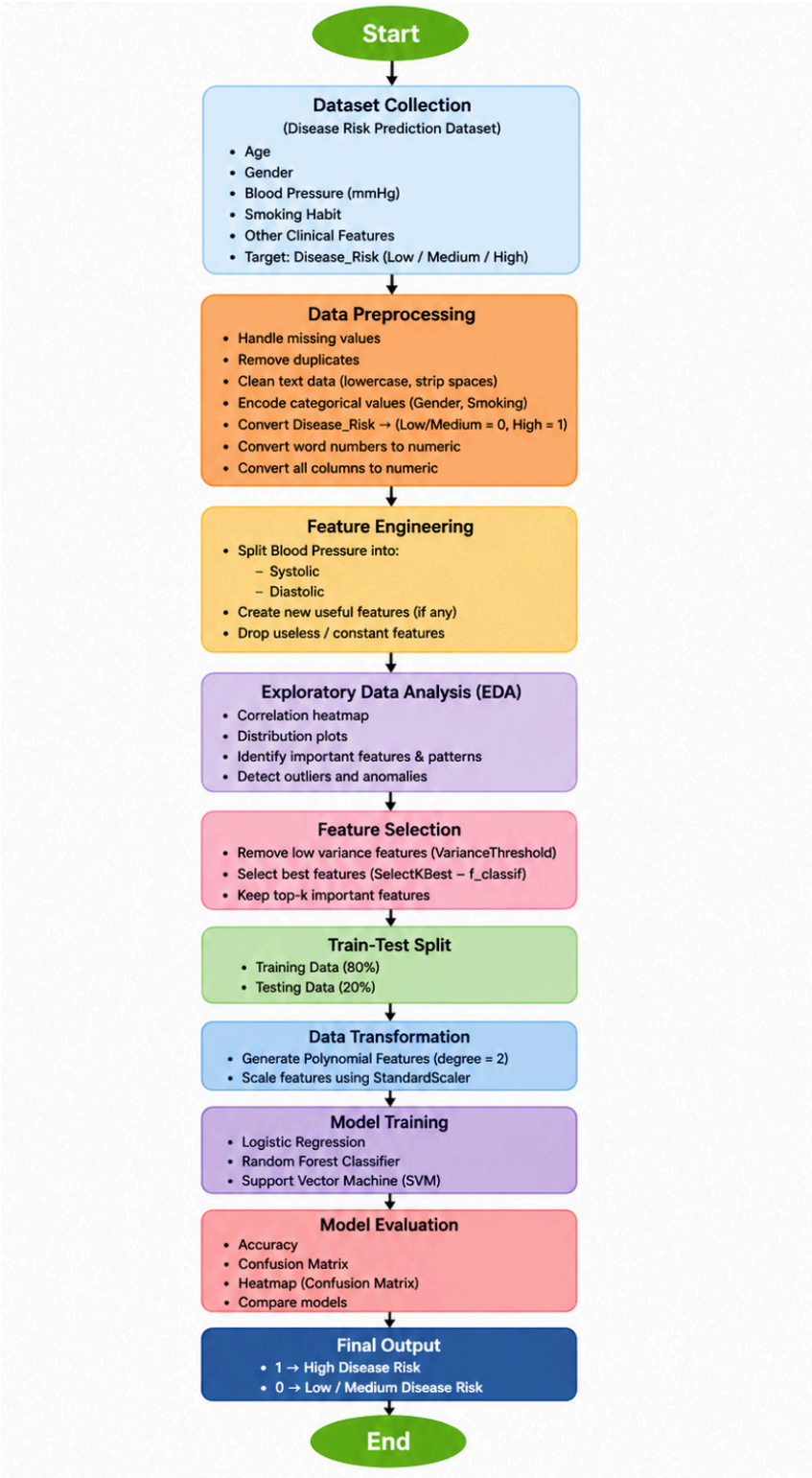

Fig 1: Showing architecture flowchart

- **Models Used(Applied/ Planned) Give a brief introduction of the models used**

Logistic Regression
A simple and efficient classification algorithm used for binary classification. It predicts the probability of a person being at high disease risk using a sigmoid function.

Random Forest Classifier
An ensemble model that combines multiple decision trees to improve accuracy and reduce overfitting. It is effective in handling complex relationships in healthcare data.

Support Vector Machine (SVM)
A powerful algorithm that finds the optimal boundary between classes. It performs well with high-dimensional datasets and can model non-linear relationships using kernels.

- **Evaluation metrics used**

1. **1. Accuracy**
2. Accuracy measures the overall performance of the model by calculating how many predictions were correct out of the total predictions made. It gives a quick understanding of how well the model is performing on the dataset.
3. However, accuracy alone may not always be sufficient, especially when the dataset is imbalanced, as it does not reflect the type of errors made by the model.
4. **2. Confusion Matrix**
5. A confusion matrix is a detailed performance evaluation tool that shows how well the model is classifying each category. It breaks down predictions into four important components:
6. **True Positive (TP):** Correctly predicted high disease risk
7. **True Negative (TN):** Correctly predicted low disease risk
8. **False Positive (FP):** Incorrectly predicted high risk when it is actually low
9. **False Negative (FN):** Incorrectly predicted low risk when it is actually high
10. This metric is especially important in healthcare applications, where certain errors (like missing a high-risk patient) can have serious consequences.
11. **3. Heatmap (Correlation Matrix)**
12. A heatmap is used to visualize relationships between different features in the dataset using color variations. It helps in understanding how strongly different variables are related to each other.
13. Identifies important features influencing disease risk
14. Detects highly correlated or redundant features
15. Helps in improving feature selection and model performance
16. **4. Model-wise Comparison**
17. Multiple machine learning models are trained and evaluated using the same dataset. Their results are compared to determine which model performs best.
18. Helps in selecting the most accurate and reliable model
19. Provides insight into how different algorithms behave on the dataset
20. Supports better decision-making for final model selection

- **Preliminary Results and Visualizations**

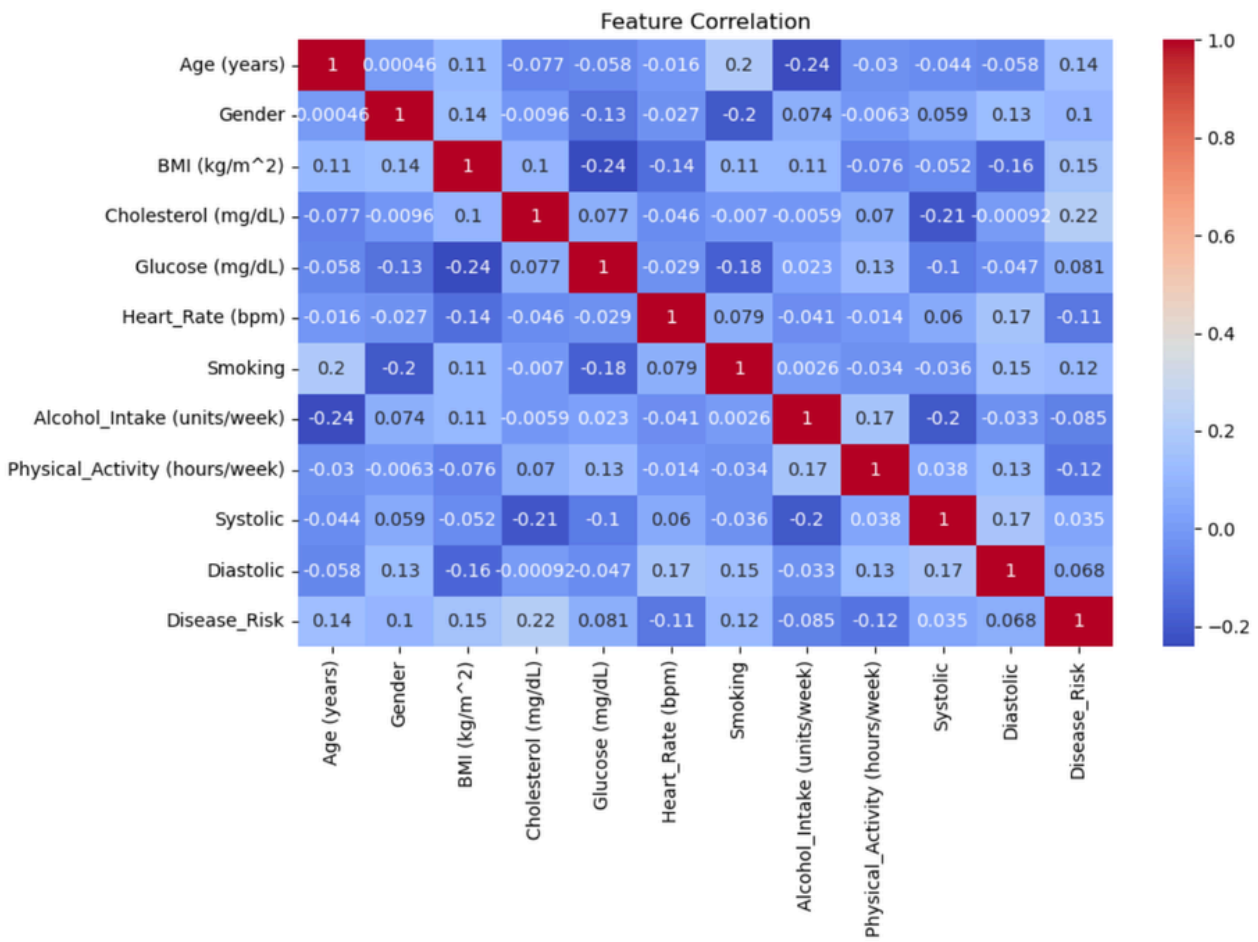

Fig 2: Figure Showing Heatmap of Feature Correlation

- ## References

1. Hands-On Machine Learning with Scikit-Learn, Keras & TensorFlow

By - Aurélien Géron (Book)

A comprehensive book covering the complete ML pipeline including preprocessing, EDA, feature selection, model building, and evaluation. Useful for understanding end-to-end system design.

2. Scikit-learn Official Documentation (ML Models & Preprocessing)

https://scikit-learn.org/stable/

Covers data preprocessing, feature selection methods (PCA), and classification algorithms like Logistic Regression, SVM, KNN, etc.

3. Kaggle Learn (Practical ML & EDA Tutorials)

https://www.kaggle.com/learn

Offers hands-on tutorials on data preprocessing, EDA, feature engineering, and machine learning workflows.

4. Official Seaborn Heatmap Documentation

https://seaborn.pydata.org/generated/seaborn.heatmap.html

Explains how to use heatmaps in seaborn library